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Nudging Students to Success: Investigating the Impact of Educational Nudges on Student Engagement and Outcomes

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Abstract

Student engagement is a key predictor of academic achievement and retention in higher education. This study investigated the impact of personalised behavioural nudges, delivered through structured phone calls and supported by personalised email/Microsoft Teams messaging, on student engagement with a Learning Management System (LMS) and academic outcomes among 192 at-risk STEM undergraduates identified by initial low LMS activity. Students received up to five phone call nudges from an academic buddy, weekly, over an eight-week period, coupled with personalised email/MS Teams follow-ups, designed to provide informational and relational support. Results showed a significant improvement in LMS engagement ($p < 0.01$) among students who responded to at least one phone call nudge compared to non-responders. Although LMS engagement was not significantly associated with module outcomes within the sample, academic outcomes, measured by module results, improved for students who engaged with the nudges; improvements were statistically significant for those who responded to two, three, and five phone call nudges ($p < 0.05$) compared to non-responders, with the highest gains observed in the five (highest nr. nudges) response group. Content analysis of student feedback highlighted four themes: Valuing Supportive Journeying with an Academic Buddy, Improved Academic Engagement, Motivation Triggered by Intervention, and wider Challenges. Findings suggest that while an initial personalised telephone call can enhance student engagement with their studies, achieving measurable academic success requires sustained intervention. This study underscores the value of relationally grounded nudges in promoting success among at-risk students.

Keywords: nudge intervention; student engagement; student outcomes; phone call nudge



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1. Introduction

Higher education (HE) is facing growing concerns with student attrition (Fahd et al., 2022) among undergraduates (Tarmizi et al., 2019), especially with first-year students who fail to continue into their second year of study (Beer & Lawson, 2017). Whilst there are a number of reasons for student attrition (Thompson et al., 2025), lack of student engagement with their studies and multiple assessment failures are among the most common (Tarmizi et al., 2019). In STEM disciplines specifically, students also face unique systemic challenges such as limited resources and insufficient institutional readiness that hinder sustained engagement and academic progression (Rehman et al., 2025). Addressing these structural gaps requires targeted interventions that are designed to bridge the disconnect between current support systems and evolving student needs. Within England (UK), the concern

with this attrition rate is reflected in the regulatory response taken by the Office for Students (OfS), which has introduced minimum expectations (thresholds) on HE institutions in relation to Continuation and Completion metrics (Condition B3).

Since students' engagement with their studies is strongly linked to academic success (Gunuc, 2014; Krause & Coates, 2008; Bowden et al., 2021), research has begun to explore the value of educational nudges as a mechanism to re-engage students with their studies (Brown et al., 2023a).

1.1. Review of Educational Nudges

Educational nudges are defined as subtle changes that guide an individual's behaviour to make decisions in a predictable way, without limiting their freedom or removing any available options (Thaler & Sunstein, 2008; Sunstein, 2015; Weijers et al., 2025). The appeal to HE institutions of an educational nudge is the relative ease of and low-cost options for implementation (Thaler & Sunstein, 2008; Garbers et al., 2023). Sunstein et al. (2018) reported a general preference for educational nudges as they target deliberate and conscious decision processes, with positive results achieved through improving active decision-making by including new information or support to the decision environment (Damgaard & Nielsen, 2018). Nudge theory recognises that changing people's behaviour in an expected way could arise from using simple nudge approaches to improve individual decision-making, as people do not always innately act in ways that best serve their own interests. Nudges, within an educational context, have the potential to trigger behaviour change by allowing students to rely on their own ability to act appropriately on the information given (i.e., content contained within the nudge), because remembering to implement an action involves many cognitive processes (Motz et al., 2021). Nudges which deliver relevant information (i.e., a task) can play a pivotal role in reducing the cognitive load of students' understanding of what is required for engagement in educational activities (Plak et al., 2023). Nudge strategies aimed at increasing student engagement often involve reframing tasks, introducing interim deadlines, and setting individualised goals to influence the cognitive processes that drive behavioural change (Garbers et al., 2023). However, the effectiveness of these interventions may be dependent on the delivery method. For instance, Weijers et al. (2025) argue that nudges can create a "messenger effect" by enhancing the persuasiveness of messages delivered by a credible and influential messenger (e.g., a teacher). They suggest that such nudges produce a larger behavioural change effect among students. This contrasts with studies that have examined motivational messages delivered via Short Message Service (SMS) about class outcomes and have generally found SMS to be an ineffective tool for nudging students toward improved academic achievement (Brandt et al., 2024).

Attempts to engage students in their studies most commonly utilise email as a reminder for attendance, assessment deadlines, and engagement with course resources. But results have not always been positive, with some interventions showing little or no effect (e.g., Oreopoulos, 2020; Ilie et al., 2022).

Zavaleta Bernuy et al. (2023) investigated the impact of email nudges sent fortnightly to improve student engagement through recording several measures to promote seminar attendance, seminar engagement, and engagement on the online learning platform. Results showed no significant improvement in engagement or overall academic achievement. This may be attributed to relatively infrequent messaging sent, too low to nudge behaviour change, and/or the low number of students who open emails and/or access email links; the high volume of emails students receive can act as a barrier to identifying key information/emails (Timmis et al., 2022). Emails may not be the most appropriate method of communicating key, time-sensitive information with the 'modern-day' student, who

may be more responsive to information originating from social media applications or text messaging (often termed nudges, [Graham et al., 2017](#)).

1.2. Summary of Nudge Research

In the educational setting, nudges given to students are an effective approach to improve awareness of key information ([Castleman & Meyer, 2020](#); [Brandt et al., 2024](#)) and have been shown to improve classroom attendance ([Graham et al., 2017](#)), student outcomes and learning behaviour linked with planning, persistence and retention ([Brown et al., 2023b](#); [Weijers et al., 2021](#)). Previous research has utilised nudges via a range of approaches, including text messaging, mobile phone apps and announcements via online learning platforms. In a study to improve academic outcomes for first-year college students with low engagement with their studies, students received text messages one to four times a month with prompts including meeting with an academic advisor, tutoring availability and course registration for the next semester. The nudge (intervention) group completed ~0.4 additional credits during the autumn semester and ~0.9 additional credits in the spring semester, with a significant increase in autumn grade point average (GPA) ([Castleman & Meyer, 2020](#)). [Blumenstein et al. \(2018\)](#) reported the use of a mobile phone app to send personalised nudges, which included study advice and academic guidance to students, resulting in improved academic performance. [Lawrence et al. \(2021\)](#) demonstrated that in a group of students who had not engaged with key course material, nudges posted on the students' Learning Management System noticeboard as 'NEWS' announcements or messages to signpost students to resources or activities to focus on for the week resulted in 5–20% increased student engagement.

Although [Bar-Gill and Cohen \(2021\)](#) have highlighted the value of low-cost (both in terms of cost and time) text message nudges compared to phone call interventions, such an approach is limited to delivering content (information), without the opportunity for the recipient to seek support or guidance on the content contained within the message ([Oreopoulos, 2020](#)). These opportunities to hold conversations between university staff and students have been highlighted as particularly valuable in supporting students, particularly those starting in HE ([Timmis et al., 2024](#)), with [Oreopoulos \(2020\)](#) demonstrating that such an approach increased graduation rates by 4% for low-engagement students who normally fail to complete their course, in addition to promoting subjective measures including wellbeing.

1.3. Nudge Behaviour Linked to Concept of Mattering

One relevant theory which underpins the impact of nudges is that of mattering ([Jones & Bell, 2025](#); [Brown et al., 2023a](#)). When students feel that they matter—by feeling important and having their contributions valued ([Prilleltensky, 2020](#)), it promotes a stronger sense of belonging ([Timmis et al., 2024](#)) and has been shown to positively impact their mental health ([G. L. Flett, 2022](#)). This is supported by studies which show that the adverse effects of long periods of physical isolation during the COVID-19 pandemic, in addition to the reliance on virtual connections, underscored the importance of relationality and engagement ([Gravett et al., 2024](#)), and addressing the need to matter significantly contributes to the health and psychological wellbeing ([Prilleltensky, 2020](#)) of post-COVID-19 university students ([G. Flett et al., 2019](#)). [G. Flett et al. \(2019\)](#) emphasise that due to the multicultural diversity prevalent in many HE institutions, there will be students who may feel marginalised and disconnected, leading to a heightened need for experiences that affirm their sense of mattering. To elaborate, [Prilleltensky et al. \(2020\)](#) indicate that HE institutions that have developed a culture by design where students experience high engagement and a sense of

matter outperform institutions with lower engagement levels, have greater achievement in various performance indicators and show superiority in goal attainment.

1.4. Study Context and Rationale

The current study examines nudging interventions designed to promote undergraduate students' engagement with their Learning Management System (LMS) and explores the associations between their engagement with these nudges, broader LMS activity, and academic outcomes. The study also draws on self-determination theory (SDT) (Deci & Ryan, 1985), which posits that motivation is enhanced when individuals experience autonomy, competence, and relatedness. Nudge interventions, when designed to support rather than control, can reinforce these psychological needs by offering timely encouragement, relevant information, and relational support.

This study extends previous research in the area through combining personalised telephone calls and more instantaneous nudges (compared to email-based approaches) via Microsoft (MS) Teams (Microsoft Corporation, Redmond, WA, USA) messaging, delivered through an 'academic buddy'—a third-space practitioner (Whitchurch, 2015; McIntosh & Nutt, 2022)—who operates across traditional academic and professional boundaries to provide holistic student support. This blended approach aims to enhance both the informational and relational dimensions of engagement.

We further extend previous work by examining the associations between the frequency with which students engage with the nudges and their broader behavioural patterns and academic outcomes.

2. Materials and Methods

2.1. Participants

192 undergraduate students (male 92%) aged 18–44 years (20 ± 3.29 yrs) from across STEM courses across the 2024/25 academic year were involved. The sample comprised:

- 73 students from year 1
- 91 students from year 2
- 28 students from year 3

Students were selected for intervention if their LMS engagement activity was less than 120 min (2 h) by Teaching Week (TW) 2—from initially inspecting the LMS engagement data of students across a range of modules, there was a clear threshold (2 h) with which to distinguish between students who were engaged and not engaged with their studies by this point in the teaching semester.

2.2. Procedure

The nudge intervention was implemented over an 8-week period, operating across two STEM Schools. In School 1, the intervention ran from TW 2 to TW 10 of the teaching semester, while in School 2, it ran from TW 3 to TW 11 (Figure 1). The implementation of intervention in TW 3 (School 2) was deliberate, to ensure sufficient capacity to complete all calls within the same School within that teaching week.

Framed as a behavioural nudge-plus intervention (Banerjee & John, 2024), the approach comprised non-coercive, personalised phone call reminders designed to increase salience and support voluntary behaviour change. While extending beyond minimal, low-touch choice architecture, the intervention intentionally integrated a relational, high-touch component. This human element aligns with emerging understandings of the added effectiveness of relational nudging (Gallego et al., 2023). Grounded in principles of choice architecture—the strategic design of decision-making environments to influence behaviour (Thaler & Sunstein, 2008; Blumenstein et al., 2018; Weijers et al., 2025)—the intervention

aimed to foster student engagement through subtle, supportive cues. Initial phone calls were used to deliver both informational and relational messages, helping to build rapport and trust. These calls encouraged students to reflect on their engagement and provided a safe space to share challenges, thereby reducing stress and promoting openness. Students were also made aware of the easy access to the academic buddy (the nudger) using MS Teams messaging, which offered ongoing support between the more substantive nudges (i.e., the phone calls). They were encouraged to reach out to their academic buddy whenever they needed support. Within the literature, there has been a varied use of ‘buddy’ in an academic context. Examples include student-buddying (peer-mentoring; [Simpson et al., 2023](#)), study buddies ([Zhai & Feng, 2025](#)), and academic staff buddies ([Fewster-Young & Corcoran, 2023](#)), all of which aim to provide academic and social support to motivate students. In this study, the academic buddy, a third-space practitioner ([Whitchurch, 2015](#)), was a Student Outcomes Tutor within the faculty who was previously unknown to the students. The academic buddy was not involved with any formal teaching with the students, but instead provided broader student support.

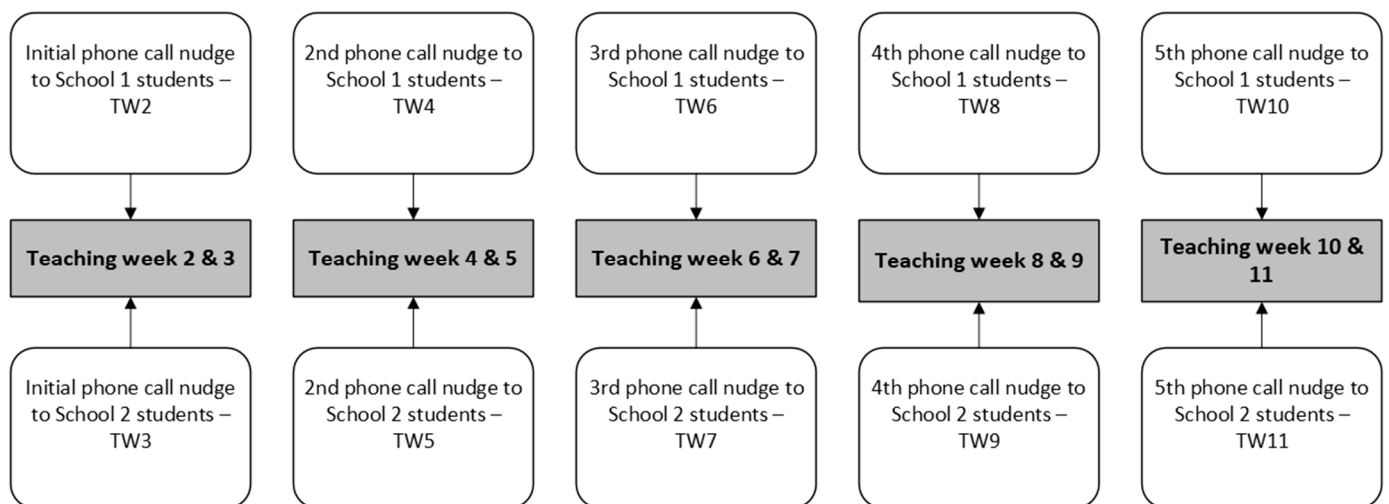


Figure 1. Timeline of phone call nudges within the academic calendar.

The phone call nudges followed a structured approach, designed to progressively support student engagement and address specific needs at different stages:

- First contact—Ascertained underlying reasons for low engagement and motivated students to engage with the LMS. The initial contact enquired about the reasons for low engagement and provided an opportunity to tailor the appropriate nudge for the individual students, e.g., developing strategies including helping students to improve organisational skills, modelling the best way for independent learning via the use of a personalised study timetable, frequent reminders for ensuring they access LMS regularly to use the uploaded resources and/or complete formative assessments, signposting to relevant support service, and motivation were all employed.
- Second contact—Praised students for any achieved progress, supported students with personal development areas such as organisational skills, and directed students to module leader or relevant support services as needed.
- Third contact—Reiterated praise for achieved progress to enhance student confidence. Students were further encouraged and motivated to maintain engagement on the LMS platform.
- Fourth and Fifth contact—Praised students for progress and signposted to appropriate support services, specifically for assistance with exam skills and subject support as they prepared for end-of-module summative assessments.

Each phone call nudge, regardless of whether it was answered, was reinforced with a follow-up email and MS Teams message to students containing the details outlined above in the first–fifth contacts. In cases where the initial call was not answered, a second attempt was made via MS Teams call later in the week. If this also went unanswered, a voice message was left to maintain contact and encourage engagement. This multi-channel approach ensured that students received timely and ongoing support, while also respecting their autonomy and availability.

2.2.1. Measures

Independent Variable:

Students were categorised into six groups based on their engagement with the nudge intervention, defined by the number of successful contacts made by the nudger (i.e., if a student answered the phone when rung):

- No response: $n = 23$
- 1 response: $n = 35$
- 2 responses: $n = 29$
- 3 responses: $n = 36$
- 4 responses: $n = 31$
- 5 responses: $n = 38$

This measure reflects students' responsiveness and availability to engage, rather than a randomly assigned or controlled intervention group-based approach, and is therefore treated as an indicator of exposure conditional on student engagement. This approach introduces a degree of selection (or survivorship) bias (Czeisler et al., 2021). Students in the “no-response” group (non-responders) did not engage in synchronous phone conversations but did receive follow-up emails, MS Teams messages, and voicemails. Consequently, this group did not represent a true control group, or non-intervention group, but rather a lower-touch comparison group receiving one-way informational contact. A true control group was not included to uphold ethical standards of equity in higher education. Following the principle that withholding all support from any student cohort is unethical (Espada-Chavarria et al., 2023), alternative designs such as standard-care or wait-list comparisons were bypassed in favour of a more inclusive methodological approach.

Dependent Variables:

Two key outcomes were measured to assess the impact of the intervention:

- LMS engagement: Total time (hours) spent on LMS—the overall time recorded from the point of module enrolment up to TW 12, which marked the end of the module. Final LMS log data were captured in TW12, at the point of the final assessment, to evaluate changes in engagement and determine the relationship between nudge frequency and platform usage.
- Academic Outcome: End-of-module grade (%)—used as a measure of academic performance. This provided a direct indicator of the potential academic impact of the nudge intervention.

Students completed two assessment components: a mid-semester summative task (multiple-choice in-class test) and an end-of-semester summative assessment (exam or coursework). The combined scores from these constituted the academic outcomes (end-of-module grade) analysed in this study.

2.2.2. Statistical Analysis

Intervention

A Kruskal–Wallis H test was conducted to analyse group main effect, with pairwise comparisons used as post hoc follow up (comparing specific group differences). Effect size was calculated using Eta squared (η^2), with the following thresholds used to denote effect size; $0.01 \leq \eta^2 < 0.06$ (small), $0.06 \leq \eta^2 < 0.14$ (medium), and $\eta^2 \geq 0.14$ (large) (Cohen, 1988; Tomczak & Tomczak, 2014).

Spearman's correlation was used to explore the relationship between LMS engagement and academic outcome.

Due to the skewed distribution of LMS hours, median-based measures were used instead of the mean to better represent the data.

2.2.3. Student Free-Text Comments

An inductive content analysis, following the approach outlined by Kyngäs (2019), was conducted on free-text comments received from students via MS Teams messages and emails. NVivo version 14 (Lumivero, Denver, CO, USA) was employed for systematic coding and thematic categorisation to identify recurring themes that reflect students' perceptions of personalised academic support delivered through the intervention. Initial open coding was applied to capture meaningful phrases and expressions without imposing predefined categories. Codes were iteratively refined and organised into four overarching themes: (a) Valuing Supportive Journeying with an Academic Buddy, (b) Improved Academic Engagement, (c) Motivation Triggered by Intervention, and (d) Challenges.

2.3. Materials

MS Teams was used to send nudge follow-up messages and make phone calls to students' personal mobile phone devices. Canvas LMS (Salt Lake City, UT, USA) analytics provided engagement log data and academic outcome records. IBM SPSS Statistics version 31.0 (Armonk, NY, USA) was utilised to perform quantitative analyses of the intervention, while NVivo version 14 was applied to facilitate content analysis of free-text comments.

3. Results

3.1. Impact of Nudge Intervention on LMS Engagement

There was a statistically significant main effect of nudge group on LMS engagement ($\chi^2(5, N = 192) = 19.95, p = 0.001, \eta^2 = 0.08$). Follow-up post hoc testing demonstrated a statistically significant increase in LMS engagement in all nudge groups ($p < 0.01$) compared to the no-response group; there was no significant difference between nudge groups one response, two responses, three responses, four responses, or five responses (Table 1).

Table 1. Comparison of each nudge group to no response group: LMS Engagement by Nudge Frequency.

Comparison	Median Difference * (h)	% Increase in LMS Engagement	Median (h)	Q1 (h)	Q3 (h)	IQR (h)
No response			1.08	0.11	6.26	6.15
No response vs. 1 response	3.37	312%	4.45	0.83	20.25	19.42
No response vs. 2 responses	2.85	264%	3.93	2.29	14.63	12.34
No response vs. 3 responses	6.65	616%	7.73	2.03	16.43	14.41
No response vs. 4 responses	4.20	389%	5.28	1.85	19.45	17.60
No response vs. 5 responses	7.05	653%	8.13	3.54	18.71	15.18

* Between-group median difference and % Increase are calculated based upon the group median increase from baseline (start of intervention) to TW12. LMS engagement ranged from 0.00 to 247.84 h. Q1—first quartile (median of the lower half); Q3—third quartile (median of the upper half), IQR—Inter-Quartile Range.

3.2. Impact of Nudge Intervention on Academic Outcomes (Module Results)

There was a statistically significant main effect of nudge group on academic outcome ($\chi^2(5, N = 192) = 13.01, p = 0.02, \eta^2 = 0.043$). Follow-up post hoc testing demonstrated a statistically significant increase in academic outcome in nudge groups two responses ($p = 0.02$), three responses ($p = 0.01$) and five responses ($p = 0.003$) compared to the no-response group (Table 2); no statistically significant difference was observed between the four-response and no-response groups ($p = 0.08$). The five-response group also achieved statistically significantly higher academic outcomes compared to the one-response group ($p = 0.02$), with a 17-point increase in module results, representing a 50% improvement.

Table 2. Comparison of nudge group 2–5 compared to no response: Academic Outcomes by Nudge Frequency.

Comparison	Difference in Module Result (%) *	% Increase in Outcomes	Median Score (%)	Q1 (%)	Q3 (%)	IQR (%)
No response			25	0	47	47
No response vs. 2 responses	21	84%	46	35	59	24
No response vs. 3 responses	20	80%	45	26	62	36
No response vs. 4 responses	10	40%	35	26	56	30
No response vs. 5 responses	26	104%	51	31	62	31

* Between-group difference calculated using median scores for each group. Q1—first quartile (median of the lower half); Q3—third quartile (median of the upper half), IQR—Inter Quartile Range. Module results ranged from 0% to 78%. For 1-response group, median score (%) = 34, Q1 (%) = 0, Q3 (%) = 55, IQR (%) = 55.

3.3. Relationship Between LMS Engagement and Academic Outcomes (Module Results)

There was no significant association between LMS engagement and module outcome ($rs(192) = 0.09, p = 0.22$).

3.4. Inductive Content Analysis and Thematic Interpretation of Student Free-Text Comments

Content analysis revealed four key themes illustrating the impact of the academic buddy initiative: (a) Valuing Supportive Journeying with Academic Buddy (coverage 38%), where students expressed appreciation for guidance and reassurance, using phrases like “thank you for your support” and “it’s really nice to have you as an academic buddy”; (b) Improved Academic Engagement (coverage 17%), reflected in comments such as “I’ve been active on Canvas [LMS]” and “will make sure to attend lectures”; (c) Motivation Triggered by Intervention (coverage 19%), highlighting the role of reminders and check-ins with comments like “promise to keep up with the class engagements” and “I will continue to engage as much as possible”; and (d) Challenges (coverage 14%), capturing barriers such as mental health struggles and course dissatisfaction, illustrated by statements like “overwhelmed due to my own incompetence” and “struggle to engage.”

4. Discussion

4.1. Impact of Phone Call Nudges on Student Engagement and Outcomes

This study examined the impact of phone call nudges on student engagement within the LMS and their subsequent academic outcomes. The findings are interpreted through an SDT lens, emphasising mattering and belonging. The findings indicate that students who responded to at least one phone call nudge demonstrated a significant improvement in LMS engagement compared to non-responders. While improved outcomes were observed for some response groups, the pattern was not strictly cumulative or proportional.

Statistically significant improvement in module results was only evident among students who responded to two, three and five phone call nudges in comparison to those who did not respond to any call, with those in the five-response group also achieving

significantly higher academic outcomes compared to the one-response group. The lack of statistical significance within the four-response group compared to the no-response group ($p = 0.08$) indicates a non-linear relationship between frequency of contact and outcomes, indicating that the results should not be interpreted as evidence of a simple dose-response effect; instead, they suggest a non-linear pattern shaped by both engagement level and assessment submissions.

Further inspection of the students within the four-response group did highlight a slightly higher proportion who did not submit any assessment components ($n = 4$; 13%), compared to the two-response ($n = 2$; 7%), three-response ($n = 3$; 8%) and five-response ($n = 2$; 5%) groups, but no difference in the rate of partial submissions across groups (four-response: $n = 3$, 10%; two-response: $n = 4$, 14%; three-response: $n = 5$, 14%; five-response: $n = 4$, 11%); suggesting that student behaviour, in terms of assessment submission, was largely similar across all groups.

Differences between groups were observed in the quality (overall grade) of submitted work, notably in the students who were awarded 30–39% in the four-response ($n = 9$; 29%), compared to two-response ($n = 4$; 14%), three-response ($n = 5$; 14%) and five-response groups ($n = 5$; 13%). This suggests that with some additional wider academic support, performance in the four-response group could improve to be in line with other nudge groups. It is also worth noting that at the current institution, students receiving a module mark between 30 and 39% are eligible to receive a ‘compensated pass’ within the module—subject to performance in associated modules.

The findings of a positive effect on LMS engagement following students who responded to at least one phone call nudge support existing research on the effectiveness of behavioural nudges as a strategy for enhancing student engagement and academic outcomes (Cortinhas, 2025). Students who received these calls reported feeling more motivated and supported, which they attributed to their increased engagement with their studies. This aligns with Yin and Wang (2016), who characterise student engagement as a multidimensional construct rooted in the dynamic interplay between motivation and engagement.

Although the intervention improved module outcomes, the results did not support the hypothesised mechanism linking academic performance to LMS engagement. Contrary to the theoretical framing in the introduction, LMS engagement was not significantly associated with module outcomes in the sample, unlike studies that showed a significant positive correlation between student engagement and academic performance (Han, 2025; Kuzminykh et al., 2021). This suggests that the intervention’s efficacy was not mediated by increased LMS platform usage, at least as captured by the engagement metrics, which did not account for the observed outcome differences and represent only a narrow behavioural indicator. Instead, the intervention may have operated through unmeasured pathways such as reduced academic anxiety, greater clarity around assessment expectations, enhanced motivation or self-efficacy, and the provision of relational and affective support. Taken together, these results indicate that within the current study, for initially low-engagement students, LMS metrics function only as a limited proxy rather than a direct mediator for authentic engagement, underscoring the need for future research to employ multimodal measures that better capture mechanisms through which supportive interventions translate into improved academic outcomes.

Collectively, the results suggest that while minimal intervention may increase engagement, to achieve a measurable academic success, this may require a higher degree of sustained behavioural change through increasingly frequent nudges. Results also suggest that to achieve significant academic improvement, at-risk students may benefit from more intensive, sustained and frequent interventions such as in-person meetings, study buddy

workshops or peer mentoring (Carragher & McGaughey, 2016), given the pivotal role that peer networks play in academic support (Wilcox et al., 2005). These findings are similarly supported by Sherr et al. (2019), who emphasised the importance of consistent nudging in promoting student success. Similarly, Azam et al. (2021) found that students who were more responsive to frequent nudges demonstrated significantly higher module scores in summative assessments, highlighting the effectiveness of repeated behavioural prompts in enhancing academic performance.

Moreover, students who had multiple successful interactions with the ‘academic buddy’ were more likely to feel motivated and supported. The perceived sense of accountability and the consistent support from an academic buddy throughout the semester appear to be key factors contributing to both increased motivation and improved performance.

4.2. Relational Support and Academic Engagement: Discussion of Key Themes from Student Comments

The academic buddy positively influenced both student experience and LMS engagement. The most notable theme—Valuing Supportive Journeying with the academic buddy—highlights the relational and emotional benefits of the intervention. In particular, the use of personalised phone call nudges, unlike other forms of educational nudges, facilitated meaningful connections with at-risk students, emerging as an effective strategy for fostering a sense of mattering and promoting inclusion (Naughton et al., 2024; Peacock & Cowan, 2019). Themes of improved academic engagement and motivation triggered by the intervention highlight the intervention’s effectiveness in promoting active learning behaviours. This aligns with evidence that proactive outreach and behavioural nudges—such as reminders and personalised check-ins—can significantly impact student engagement and completion of key academic tasks (Page et al., 2025). Conversely, the less frequent theme of challenges revealed barriers, including mental health struggles and structural constraints. The existing literature consistently demonstrates that elevated levels of depression, anxiety, and stress are major impediments to student engagement and success (Hyseni Duraku et al., 2024). The personalised phone call nudges helped mitigate these barriers by integrating academic support with mental health strategies and inclusive practices. One might argue that through dedicating resources to increase engagement among low-engagement students, this risks neglecting those who are already engaged in higher education. This reflects a broader education philosophical question, which needs to be considered against the OfS (independent regulator for HE in England, UK) expectation that all students should be given the opportunity to succeed in HE. To ensure equitable support, it is worth recognising that in the current study, the additional resource to support low-engagement students was provided through a third-space practitioner rather than an academic or module leader. The personal tutors (academic), module leaders, and teachers continue to support all students within their respective modules and tutee groups. Supporting emotional and social factors in learning is a collective institutional duty rather than the sole duty of individual teachers. Creating environments of high expectations and high support (Felten & Lambert, 2020; Gravett, 2023) should be a shared duty of care for all staff in HE. This relationship-based approach allows staff to more effectively address academic needs while empowering students towards autonomous, independent learning.

4.3. The Role of Interpersonal Nudges in Fostering Engagement and Academic Success

Beyond supporting SDT needs, the interpersonal connection cultivated through the phone call nudges played a pivotal role in strengthening students’ sense of mattering within the academic environment. This is a critical finding, given that mattering is widely acknowledged as essential for fostering student engagement, as G. L. Flett et al. (2022) highlight. They note that mattering enhances student engagement by creating a supportive,

relational learning environment that promotes confidence, motivation, and persistence. Relational support, such as that offered through the intervention, helps reduce attrition and improve academic outcomes by fostering a stronger sense of belonging (Rehman et al., 2023). Rehman et al. (2023) demonstrate that a strong sense of belonging is significantly associated with student retention and is linked to various psychological and socio-emotional benefits, including improved wellbeing, reduced stress levels, and increased self-esteem.

For students in this study, the emotional and social support provided through personalised outreach was not only beneficial for their wellbeing but also served as a strong motivator for sustained academic involvement. This is particularly relevant in STEM disciplines, where students often face unique challenges related to identity, representation, and inclusion (Singer et al., 2020).

These findings align with the concept of a recursive spiral, as described by Edwards et al. (2021) and Krause-Levy et al. (2021), where a strengthened sense of belonging leads to deeper engagement, which in turn reinforces belonging and contributes to academic success. This cyclical interplay highlights the potential of targeted, personalised interventions—such as proactive outreach via phone calls—to positively influence both student wellbeing and academic performance.

The nudges were effective not only in prompting engagement but also in delivering timely and relevant information (Lawrence et al., 2021). The fourth and fifth nudges, in particular, focused on preparing students for end-of-semester assessments and signposted them to Study Skills workshops for exam support.

The success of phone call nudges highlights the importance of timely and personalised communication in supporting student learning. As Lawrence et al. (2019) argue, effective expectation management is essential in helping students navigate academic discourse and engage meaningfully with their learning. This study suggests that empathetic and well-timed nudges can play a vital role in bridging the gap between institutional expectations and student realities, particularly in diverse and multicultural cohorts.

4.4. Limitations and Directions for Future Research

A true control group was not included in this study, nor were alternative designs such as standard-care or wait-list comparisons implemented, as the research team deemed withholding all forms of support was inconsistent with the principles of equity and equality within the HE context. Instead, the “no-response” group is presented as a lower-touch comparison condition: these students were offered the same support opportunities but did not engage with them, and therefore do not constitute a true control group. Consideration was given towards comparing outcomes between modules of previous academic years and utilising the data as a basis for a control group, but factors such as changing module leader/teacher, teaching structure and assessment design limited the validity of these comparisons. Consequently, historical data was not considered a meaningful basis for a control condition. Additionally, by selecting students with very low early engagement, it is expected that, without any intervention, natural improvements in engagement would occur during the semester. This observation, alongside self-selection and survivorship bias (Czeisler et al., 2021), reinforces that any observed improvements cannot be conclusively attributed to the intervention, as they may partly reflect differences in student propensity to engage with their studies (c.f., Hibbs et al., 2026). Whilst the internal comparisons between nudge groups partially mitigate this issue, future research should investigate this ‘natural improvement’ and the associated factors that impact such improvement.

The number of successful phone contacts was driven by student responsiveness rather than a controlled level of engagement; the results may reflect underlying behaviours of students. Students may have differed in their profiles or compliance behaviours, which

could have impacted their tendency to respond and engage. These findings should therefore be viewed as correlational as opposed to causal, with further work required to demonstrate causality. Whilst the participants involved in this research represented a diverse and multicultural sample, this was limited to students studying full-time undergraduate STEM subjects. Further research is warranted to explore how additional factors, including a wider range of subjects, in addition to understanding how factors including student demographics and type/mode of study affect the efficacy of phone call nudges.

5. Conclusions

Nudging has gained increasing attention in HE as a promising approach to influencing student behaviour and engagement, opening new avenues for research and practice (Weijers et al., 2021). The current research demonstrates that personalised phone call nudges, delivered through an ‘academic buddy’, promote behavioural change in low-engagement students. However, grounded in behavioural science, the phone call nudge intervention extends beyond traditional low-resource, nudge-based research. It adheres to core nudge principles while transcending the commonly adopted low-cost approach prevalent in the existing literature. Although the sample’s demographic profile (92% male) reflects the structural gender disparity within UK STEM disciplines (approximately 78% male in Engineering and Technology and Computing; HESA, 2025), this composition nonetheless represents a limitation and constrains the generalisability of the findings to more gender-balanced populations. Future work is needed to extend our findings across both female-dominated and gender-balanced cohorts. Through students engaging in sustained nudges, it has been possible to observe an improvement in their outcome; those students who engaged in two or more nudges significantly improved their outcome compared to students who only responded to one or no nudges. The intentional early timing of the intervention within the academic cycle likely contributed to the successes observed.

Overall, these results highlight the substantial potential of personalised and relational nudging interventions as a scalable and impactful strategy for improving student outcome metrics.

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Abbreviations

The following abbreviations are used in this manuscript:

LMS	Learning Management System
MS	Microsoft
STEM	Science, Technology, Engineering, and Mathematics

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